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COURSE: COMPUTER NETWORKS

LAB: 03

Task 1:

What is Ethernet cable? What type of Ethernet cable is most commonly used in LAN network?

An Ethernet cable is a type of network cable used to connect devices in a wired local area network (LAN), such as computers, routers, and switches, to transmit data.

The most commonly used Ethernet cable in LAN networks is the Category 6 (Cat6) cable. It supports speeds up to 10 Gbps and is widely used for both home and office networks due to its performance and affordability.

Task 2:

What is a LAN (local area network)? When and why use a LAN cable to send files from one computer to another?

A LAN (Local Area Network) is a network that connects devices within a limited area, such as a home, office, or building, allowing them to share resources and communicate.

When and why to use a LAN cable:

- Use a LAN cable (Ethernet cable) when you need fast, reliable, and secure file transfers.
- It is ideal for transferring large files (e.g., videos, backups) because it offers higher speeds and lower latency compared to Wi-Fi.
- It is more secure than wireless transfers, as data is not broadcast over the air.

Task 3:

How Does LAN Work? What is meant by 127.0.0.1 and localhost?

Devices in a LAN are connected via switches or routers using Ethernet cables or Wi-Fi. Each device has a unique IP address. When one device wants to communicate with another, it sends data packets addressed to the recipient's IP or MAC address. The switch or router directs these packets to the correct device.

192.20.0.1 and localhost:

- 192.20.0.1 is a special IP address called the loopback address. It refers to the device itself.
- localhost is a hostname that also points to the same loopback address (192.20.0.1).
- Both are used to test network software on the same device without needing an external network.

Task 4:

How two computers can share files through disk sharing

1. Connect the computers to the same network (via a router/switch or directly using a crossover Ethernet cable).
2. Enable disk sharing on the computer containing the files:
 - Go to Folder Options → View → enable “Network discovery” and “File and printer sharing”.
 - Right-click the folder/drive → Properties → Sharing tab → Share or Advanced Sharing.
3. Access the shared disk from the other computer:
 - Open File Explorer → go to Network → find the other computer → enter its username/password if prompted.
 - Alternatively, type \\[IP_address_or_computer_name] in the address bar.
4. Copy or move files between the shared disk and the local computer.

This allows direct file transfer without external devices or internet.

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